

Inversion method:

Let $u \sim U[0, 1]$, $X \sim F(x)$, where $F^{-1}(\cdot)$ exists

then $F^{-1}(u) \sim F(x)$

proof:
$$P(F^{-1}(u) \leq x) = P(u \leq F(x))$$
$$= F(x)$$

Example: generate exponential distribution

$$f_X(x) = \lambda e^{-\lambda x}$$

$$\begin{aligned} F(x) &= \lambda \int_0^x e^{-\lambda z} dz \\ &= \lambda \left(-\frac{1}{\lambda} e^{-\lambda z} \Big|_{z=0}^x \right) = \lambda \left(\frac{1}{\lambda} - \frac{1}{\lambda} e^{-\lambda x} \right) \\ &= 1 - e^{-\lambda x} \end{aligned}$$

$$\begin{aligned} \text{Let } y &= 1 - \exp(-\lambda x) \\ \exp(-\lambda x) &= 1 - y \Rightarrow -\lambda x = \log(1-y) \Rightarrow x = \frac{\log(1-y)}{-\lambda} \\ &\Rightarrow F^{-1}(x) = \frac{\log(1-x)}{-\lambda} \end{aligned}$$

$$\text{So } \frac{\log(1-u)}{-\lambda} \sim \text{exp}(\lambda)$$

problem: not all distribution has $F^{-1}(\cdot)$

e.g. Gaussian distribution has no explicit form for $F^{-1}(\cdot)$

Acceptance - Rejection method

Suppose we have a density function f we cannot directly sample from, but we can sample from another density function g , such that there exist a $C < \infty$

$$f(x) \leq Cg(x), \quad \forall x$$

then acceptance-rejection method do

While True:

$$X \sim g(x)$$

$$u \sim U[0, 1]$$

$$\text{if } u \leq \frac{f(x)}{Cg(x)}:$$

accept X

proof: $p(X \leq z \mid u \leq \frac{f(x)}{Cg(x)})$

$$= \frac{p(X \leq z, u \leq \frac{f(x)}{Cg(x)})}{p(u \leq \frac{f(x)}{Cg(x)})}$$

$$= \frac{\int_{-\infty}^z \int_0^{\frac{f(x)}{Cg(x)}} g(x) du dx}{\int_{-\infty}^{+\infty} \int_0^{\frac{f(x)}{Cg(x)}} g(x) du dx}$$

$$= \frac{\int_{-\infty}^z \frac{f(x)}{C} dx}{\int_{-\infty}^{+\infty} \frac{f(x)}{C} dx}$$

$$= \frac{\int_{-\infty}^z f(x) dx}{\int_{-\infty}^{+\infty} f(x) dx}$$

$$= \int_{-\infty}^z f(x) dx = F_X(z)$$

Box-Muller method

$$\text{Let } \theta \sim U[0, 2\pi], \quad R^2 \sim \exp(1/2)$$

$$\text{and } X = R \cos \theta, \quad Y = R \sin \theta$$

$$\text{then } X, Y \sim N(0, 1)$$

proof: Denote $f\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\text{Since } X = R \cos \theta, \quad Y = R \sin \theta$$

$$\Rightarrow X^2 + Y^2 = R^2$$

$$\Rightarrow R = \sqrt{X^2 + Y^2} = (X^2 + Y^2)^{\frac{1}{2}}$$

$$\text{Also, } \frac{Y}{X} = \tan \theta \Rightarrow \theta = \tan^{-1}\left(\frac{Y}{X}\right)$$

$$\Rightarrow f^{\text{tr}} \text{ exists, and Jacobian} = \begin{pmatrix} x(x^2+y^2)^{-\frac{1}{2}} & y(x^2+y^2)^{-\frac{1}{2}} \\ -\frac{1}{1+(y/x)^2} \cdot \frac{y}{x^2} & \frac{1}{1+(y/x)^2} \cdot \frac{1}{x} \end{pmatrix}$$

$$\begin{aligned} \Rightarrow |J| &= \frac{1}{1+(y/x)^2} \cdot \frac{1}{\sqrt{x^2+y^2}} + \frac{y}{x^2+y^2} \cdot \frac{1}{\sqrt{x^2+y^2}} \\ &= \frac{1}{\sqrt{x^2+y^2}} \left(\frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2} \right) \\ &= \frac{1}{\sqrt{x^2+y^2}} \end{aligned}$$

$$\begin{aligned} \Rightarrow f_{X,Y}(x,y) &= \frac{1}{2\pi} \sqrt{x^2+y^2} \exp\left(-\frac{1}{2}(x^2+y^2)\right) \cdot \frac{1}{\sqrt{x^2+y^2}} \\ &= \frac{1}{2\pi} \exp\left(-\frac{1}{2}(x^2+y^2)\right) \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right) \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2\right) \end{aligned}$$

$$\hookrightarrow \text{if } R^2 \sim \exp(1/2)$$

$$\text{then } P(R < x)$$

$$= P(R^2 < x^2)$$

$$= \int_0^{x^2} \frac{1}{2} e^{-\frac{1}{2}r} dr$$

$$= \frac{1}{2} (-2 e^{-\frac{1}{2}r}) \Big|_{r=0}^{x^2}$$

$$= 1 - e^{-\frac{1}{2}x^2}$$

$$f_R(R=r) = (-e^{-\frac{1}{2}r^2}) (-\frac{1}{2} \cdot 2r)$$

$$= r e^{-\frac{1}{2}r^2}$$